

6

Spot Price Models and Pricing Standard Instruments

6.1 Introduction

In this chapter we consider models for pricing energy derivatives which are formulated in terms of the spot energy price. We investigate a number of models for the stochastic behavior of energy prices and show how futures/forward contracts and standard European options can be priced with analytical formulae. The models we look at range from the single factor Black (1976) model to a three-factor model with stochastic convenience yield and stochastic term structure of interest rates. We also comment on generalising the models to include seasonal factors throughout, handling volatility smiles is discussed in chapter 7. The numerical techniques required to price more complex energy derivatives are the subject of chapter 7.

6.2 Single Factor Models

Probably the best known assumption governing the dynamics of a commodity or energy spot price is given by the well known stochastic differential equation (1.7) which we repeat here;

$$dS = (r - \delta)Sdt + \sigma Sdz \quad (6.1)$$

Equation (6.1) represents the risk-neutral process for the spot energy price, S , where r and δ are constants representing the short term interest rate and convenience yield respectively, and where σ is the volatility of proportional changes in the spot energy price. The concept of a convenience yield, viewed as an implicit net dividend yield accruing to the owner of a commodity at the margin is now a well understood concept in the commodity futures literature¹.

The value of any contingent claim on the spot price, C , under the dynamics assumed in equation (6.1) must satisfy the following well-known partial differential equation;

$$\frac{1}{2}\sigma^2 S^2 C_{SS} + (r - \delta)SC_S + C_t - rC = 0 \quad (6.2)$$

where subscripts denote partial derivatives.

¹ See for example Hull (2000).

Futures and Futures Pricing

Black (1976) showed that under this process the futures price, with a boundary condition equating the futures price to the spot price at maturity, s , satisfies the cost of carry relationship;

$$F(t, s) = Se^{(r-\delta)(s-t)} \quad (6.3)$$

As interest rates are constant, this is also the price of a forward contract with the same specifications (see Cox and Ross (1976) for the equivalence of futures and forwards when interest rates are constant). Figure 6.1 shows how the futures curve is influenced by the convenience yield. Futures prices are calculated via equation (6.3) with an original spot price of 100 and a short rate of 5% with typical values for the convenience yield of 1%, 5% and 10%. For low convenience yield ($\delta > r$) of 1% the forward curve increases exponentially at the interest rate less the convenience yield i.e. 4% per year, a convenience yield of 5% equal to the interest rate ($\delta = r$) gives a flat forward curve and a high convenience yield ($\delta < r$) of 10% gives an exponentially declining forward curve. The simple shape of these forward curves is clearly not realistic but the model can easily be generalised to allow the market forward prices to be used, we discuss this later in this chapter and also in chapter 7.

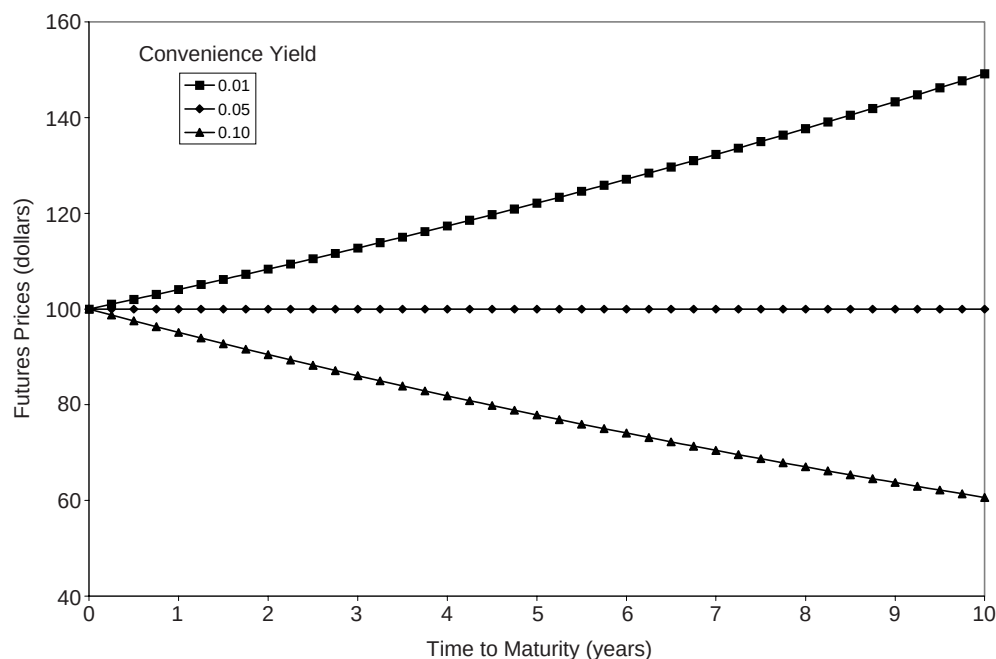


FIGURE 6.1 Futures Prices from Cost-of-Carry for Typical Convenience Yields

A straightforward application of Itô's lemma can be used to show that the volatilities of the futures prices are constant and equal to the volatility of the spot price, i.e.;

$$\sigma_F(t, s) = \sigma \quad (6.4)$$

where $\sigma_F(t, s)$ is the volatility of the s -maturity futures price. We have already seen that forward/futures price volatilities vary with maturity, typically declining relative to the spot price volatility for increasing maturities and also often exhibiting seasonal patterns. We show in the following sections that the model's description of forward price volatilities can be improved by the introduction of mean reversion and time varying spot volatility.

Option Pricing

Black (1976) also developed a pricing formula for valuing European futures options. Let K represent the strike price of the option, the value of a European call option at time t , maturing at time T on a futures contract that matures at time s is given by;

$$\begin{aligned} c(t, T, s) &= P(t, T)[F(t, s)N(h) - KN(h - \sigma\sqrt{T - t})] \\ h &= \frac{\ln(F(t, s)/K) + \frac{1}{2}\sigma^2(T - t)}{\sigma\sqrt{T - t}} \end{aligned} \quad (6.5)$$

In the strict interpretation of the Black model, $F(t, s)$ is the analytical forward price given by equation (6.3) with constant volatility σ and $P(t, T)$ represents the T -maturity discount factor. In practice this can be replaced by the market price of the forward/futures contract – this corresponds to making the convenience yield a general function of time in order to make the spot price process consistent with the current market forward prices. This technique is very general and we show in this chapter and chapter 7 how spot price models can usually be made consistent with the market forward curve in this way. The spot price volatility can also be made time varying to allow for the modelling of seasonal volatility patterns. We describe how this can be done for most spot price models beginning with the Schwartz model which is discussed in the next section. Figure 6.2 illustrates the effect of volatility on the price of a European call futures option. The option is assumed to have a maturity of 3 months with an at-the money strike and original futures price of 100. We plot the European call prices for volatilities of 10%, 25% and 50%.

The Schwartz Single Factor Model

As we have already noted, the constant volatility assumption of the Black model is not consistent with the empirical observation that longer dated forwards are less volatile than short dated forwards. A more realistic single factor model was introduced by Schwartz (1997) in which he assumes that the spot price follows a mean reverting type

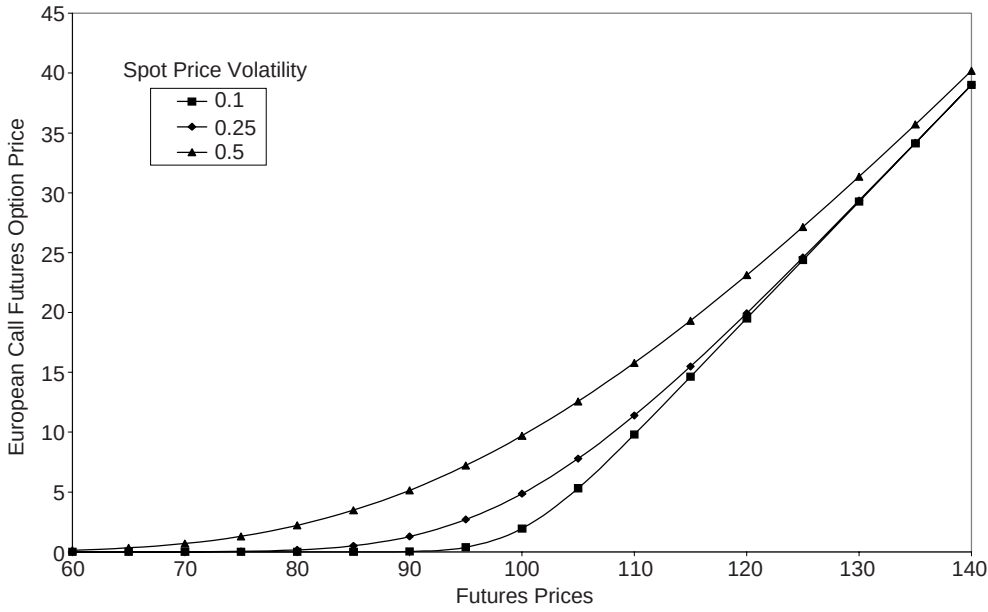


FIGURE 6.2 Option Prices in the Black (1976) Model for Typical Volatilities

process (see chapter 2 for a discussion of simulating mean reverting spot price behaviour and chapters 2 and 3 for a discussion of parameter estimation):

$$dS = \alpha(\mu - \lambda - \ln S)Sdt + \sigma Sdz \quad (6.6)$$

where α is the mean reversion rate – the speed of adjustment of the spot price back towards its long term level μ , σ is the spot price volatility and λ the market price of energy risk.

Defining $x = \ln S$ and applying Itô's lemma to equation (6.6), the log price can be characterised by the Ornstein-Uhlenbeck process;

$$dx = \alpha(\hat{\mu} - x)dt + \sigma dz \quad (6.7)$$

where $\hat{\mu} = \mu - \lambda - \frac{\sigma^2}{2\alpha}$.

This process leads to the following differential equation for all contingent claims whose payoff depends on the level of the energy spot price and time;

$$\frac{1}{2}\sigma^2 S^2 C_{SS} + \alpha(\mu - \lambda - \ln S)SC_S + C_t - rC = 0 \quad (6.8)$$

Futures and Forward Pricing

With the appropriate boundary conditions, futures and forward prices with maturity s are equal and given by;

$$F(t, s) = \exp \left[e^{-\alpha(s-t)} \ln S + (1 - e^{-\alpha(s-t)}) \left(\mu - \lambda - \frac{\sigma^2}{2\alpha} \right) + \frac{\sigma^2}{4\alpha} (1 - e^{-2\alpha(s-t)}) \right] \quad (6.9)$$

Figure 6.3 shows the sensitivity of the futures price in equation (6.9) to varying the speed of mean reversion. The parameters used for the figure are $S = 110$, $\sigma = 0.3$, and $\lambda = 0$ with μ fixed at $\ln(100)$ – thus the current spot price is \$10 above the long term level μ . The mean reversion rate determines how soon, in terms of maturity, forward prices stabilise at their long maturity level. Notice that the long maturity level of the futures curve is not equal to $\exp(\mu)$ but is adjusted by an amount which depends on the relative size of α and σ ;

$$F(t, \infty) = \exp \left[\mu - \lambda - \frac{\sigma^2}{4\alpha} \right] \quad (6.10)$$

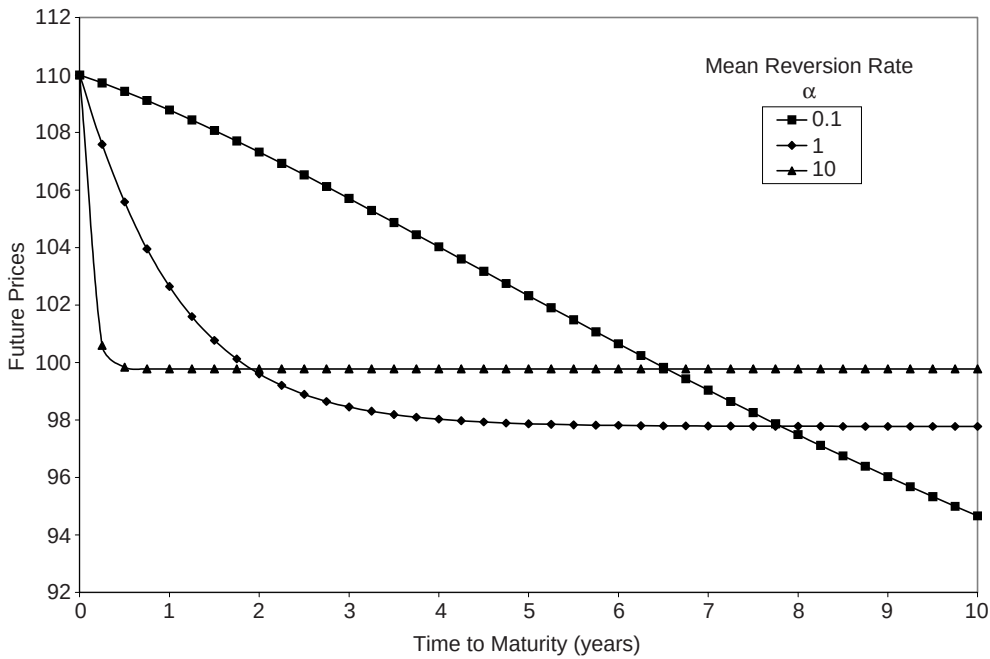


FIGURE 6.3 Futures Prices in the Schwartz Single Factor Model for Typical Mean Reversion Rates

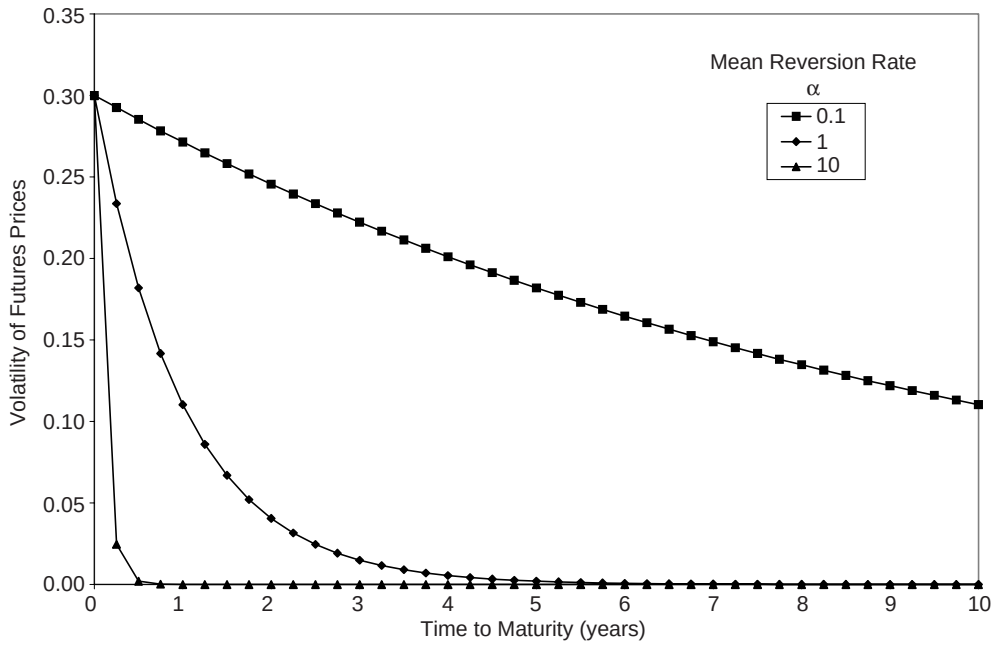


FIGURE 6.4 Volatility of Futures Prices in the Schwartz Single Factor Model for Typical Mean Reversion Rates ($\sigma = 0.3$)

Itô's lemma can be applied to equation (6.9) to determine the term structure of proportional futures volatilities in this single factor model, yielding:

$$\sigma_F(t, s) = \sigma e^{-\alpha(s-t)} \quad (6.11)$$

Figure 6.4 graphs the effect that the speed of mean reversion has on the term structure of volatility of futures prices. Increasing the speed of mean reversion steepens the attenuation of the volatility curve. Equation (6.11) and Figure 6.4 illustrate that as the maturity of the forward contract approaches large values the volatility of the contract tends to zero. This attenuation of volatility increases with increasing speed of mean reversion.

The volatility structure under the Schwartz single factor model is more realistic than the Black model but still has quite a simple shape. In particular the volatilities tend to zero for longer maturities and this happens for maturities less than one year for mean reversion rates larger than about 7. Although the market volatilities of forward prices decline with maturity they never get close to zero and so the Schwartz model has a problem for pricing options on long maturity forward contracts. We will see in chapter 8 how a volatility function of similar form is able to capture the attenuation of typical market forward volatility structures.

Option Pricing

Clelland and Strickland (1999a)² show that for the one factor Schwartz model, the price of a European call option maturing at time T on a futures contract that matures at time s is given by;

$$\begin{aligned} c(t, T, s) &= P(t, T)[F(t, s)N(h) - KN(h - \sqrt{w})] \\ h &= \frac{\ln(F(t, s)/K) + \frac{1}{2}w}{\sqrt{w}} \end{aligned} \quad (6.12)$$

In equation (6.12) w^2 is given by the integral of the forward return variance over the life of the option:

$$\begin{aligned} w^2 &= \int_t^T \sigma^2 e^{-2\alpha(s-u)} du \\ &= \frac{\sigma^2}{2\alpha} \left(e^{-2\alpha(s-T)} - e^{-2\alpha(s-t)} \right) \end{aligned} \quad (6.13)$$

If $s = T$ then the option on a future becomes an option on the spot energy and so the price of a T -maturity option on the spot price $c(S, t, T)$ is given by;

$$\begin{aligned} c(S, t, T) &= P(t, T)[F(t, T)N(h) - KN(h - \sqrt{w})] \\ h &= \frac{\ln(F(t, T)/K) + \frac{1}{2}w}{\sqrt{w}} \\ w^2 &= \frac{\sigma^2}{2\alpha} \left(1 - e^{-2\alpha(T-t)} \right) \end{aligned} \quad (6.14)$$

In a similar way to the Black model the analytical forward price given by equation (6.9) can be replaced by the market price in the option pricing equation allowing the model to be consistent with the market observed forward prices. This is equivalent to allowing the mean reversion level to be a general function of time (see section 7.3 for more details). Furthermore, since the option price only depends on the integral of the forward return variance over the life of the option, the parameters α and σ can be made time dependent allowing the inclusion of seasonal patterns. The only complication this introduces is that the integral in equation (6.13) may now need to be evaluated numerically, however, the option pricing equation remains the same. In chapter 8 we show how analytical volatility functions, such as equation (6.11), can be calibrated to historically estimated market volatilities or implied volatilities. The incorporation of the observable volatilities, as well as forward prices, is very important in practice because this allows relatively simple models to price options as consistently as possible with the actual behaviour of the underlying market prices.

² Related results have appeared in Jamshidian (1991), Amin and Jarrow (1991, 1992) and Amin, Ng and Pirrong (1995).

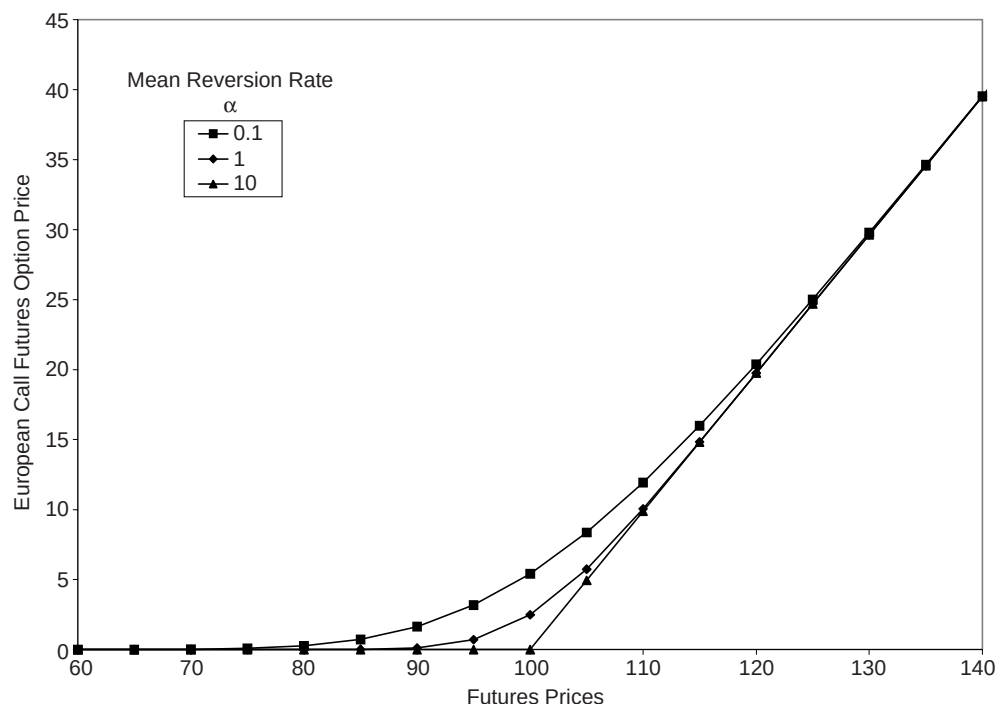


FIGURE 6.5 Option Prices in the Schwartz Single Factor Model for Typical Mean Reversion Rates

Figure 6.5 shows how the speed of mean reversion parameter affects the prices of European futures call options. The option has a maturity of 3 months on a forward contract with an original maturity of 1 year. The strike price of the option is assumed to be 100, the interest rate is 5% and the volatility is 30%. The values for α used in the figure are 0.1, 1 and 10.

Finally, for this section, we analyze the effect of the mean reverting process on option prices compared with the Black assumption of constant forward volatilities. Table 6.1 prices options with maturities of 9 months, 1 year, and 1.25 years on a forward with 1.5 years to maturity. The original forward price is assumed to equal 100, interest rates are 5% and the volatility of the spot asset is assumed to be 50%.

The Black's prices are derived from equation (6.5) and the Schwartz prices from equation (6.12) with mean reversion parameter α which we vary from 0.01 to 0.25. Table 6.1 shows that as we reduce the mean reversion parameter the Schwartz prices approach the Black prices – a parameter of $\alpha = 0$ would imply a flat term structure of volatility and the Schwartz prices equal to Black prices.

TABLE 6.1 Comparison of Option Prices in the Black (1976) and Schwartz (1997) Single Factor Models

		Black	Schwartz		
		Vary mean reversion (α)			
$T = 0.75$	K		0.01	0.1	0.25
	95	11.849	11.747	10.885	9.644
	100	9.589	9.482	8.577	7.268
	105	7.681	7.574	6.665	5.358
$T = 1$	K		0.01	0.1	0.25
	95	12.933	12.831	11.963	10.704
	100	10.789	10.682	9.777	8.458
	105	8.941	8.834	7.919	6.593
$T = 1.25$	K		0.01	0.1	0.25
	95	13.798	13.701	12.872	11.668
	100	11.754	11.652	10.791	9.536
	105	9.967	9.864	8.992	7.725

6.3 Two Factor Models

Some of the deficiencies of the single factor models, namely the very simple shape of the volatility structure and its attenuation to zero with increasing maturity can be remedied by introducing a second stochastic factor into the model. Pilipovic (1997) describes a two-factor mean-reverting model where spot prices revert to a long term equilibrium level which is itself a random variable³. Gibson and Schwartz (1990), Schwartz (1997) and Hilliard and Reis (1998) (HR) all analyze versions of the same two-factor model that allows for a stochastic convenience yield and permits a high level of analytical tractability. The first factor is the spot price process which is assumed to follow the GBM of equation (6.1). The second factor is the instantaneous convenience yield of the spot energy and is assumed to follow the mean reverting process;

$$d\delta = \alpha_{\delta}(\bar{\delta} - \delta)dt + \sigma_{\delta}dz_{\delta} \quad (6.15)$$

In equation (6.15) α_{δ} and $\bar{\delta}$ represent the speed of adjustment and long term mean of the convenience yield respectively with σ_{δ} the volatility of the convenience yield. The increments to the Brownian motions driving the spot price and convenience yield are

³ Pilipovic derives a closed-form solution for forward prices to her model when the spot and long term prices are uncorrelated, but however doesn't discuss option pricing in her two-factor framework.

assumed to have correlation coefficient $\rho_{S\delta}$. The joint process for the spot and convenience yield lead to the following differential equation for contingent claim prices;

$$\frac{1}{2}C_{SS}S^2\sigma^2 + \frac{1}{2}C_{\delta\delta}\sigma_\delta^2 + C_{S\delta}S\rho_{S\delta}\sigma\sigma_\delta + C_S S(r - \delta) + C_\delta(\alpha_\delta(\bar{\delta} - \delta) - \lambda_\delta\sigma_\delta) + C_t = 0 \quad (6.16)$$

where λ_δ is the market price of risk per unit of convenience yield. Since the convenience yield is not a tradable variable, equation (6.16) depends on investor risk preferences which are embedded in the market price of convenience yield risk, λ_δ .

Futures and Forward Pricing

With interest rates assumed to be constant, futures and forward energy prices are equal. Schwartz (1997) derives an expression for the futures price under the assumption that the interest rate term structure is flat and constant at a rate r .

$$\begin{aligned} F(t, s) &= S \exp\left(-\delta \frac{1 - e^{-\alpha_\delta(s-t)}}{\alpha_\delta} + A(t, s)\right) \\ A(t, s) &= \left(r - \hat{\delta} + \frac{1}{2} \frac{\sigma_\delta^2}{\alpha_\delta^2} - \frac{\sigma\sigma_\delta\rho_{S\delta}}{\alpha_\delta}\right)(s-t) + \frac{1}{4} \sigma_\delta^2 \frac{1 - e^{-2\alpha_\delta(s-t)}}{\alpha_\delta^3} \\ &\quad + \left(\hat{\delta}\alpha_\delta + \sigma\sigma_\delta\rho_{S\delta} - \frac{\sigma_\delta^2}{\alpha_\delta}\right) \frac{1 - e^{-\alpha_\delta(s-t)}}{\alpha_\delta^2} \\ \hat{\delta} &= \bar{\delta} - \frac{\lambda_\delta}{\alpha_\delta} \end{aligned} \quad (6.17)$$

HR, assuming the same processes as Schwartz, derive the futures pricing formula consistent with the observed term structure of interest rates, given in terms of pure discount bond prices by $P(t, s)$,

$$\begin{aligned} F(t, s) &= S e^{-H_\delta(t, s)\delta} \frac{A(t, s)}{P(t, s)} \\ A(t, s) &= \exp\left[\frac{(H_\delta(t, s) - (s-t))(\alpha_\delta^2\bar{\delta} - \alpha_\delta\lambda_\delta\sigma_\delta - \sigma_\delta^2/2 + \rho_{S\delta}\sigma\sigma_\delta\alpha_\delta)}{\alpha_\delta^2} - \frac{\sigma_\delta^2 H_\delta(t, s)^2}{4\alpha_\delta}\right] \\ H_\delta(t, s) &= \frac{1 - e^{-\alpha_\delta(s-t)}}{\alpha_\delta} \end{aligned} \quad (6.18)$$

If we set the initial term structure to be $P(t, s) = e^{-r(s-t)}$ then it is easy to verify that the futures pricing formulas given by equations (6.9) and (6.18) are equivalent. Also if we set $\sigma_\delta = \alpha_\delta = 0$ then equations (6.9) and (6.18) reduce to the cost of carry relationship,

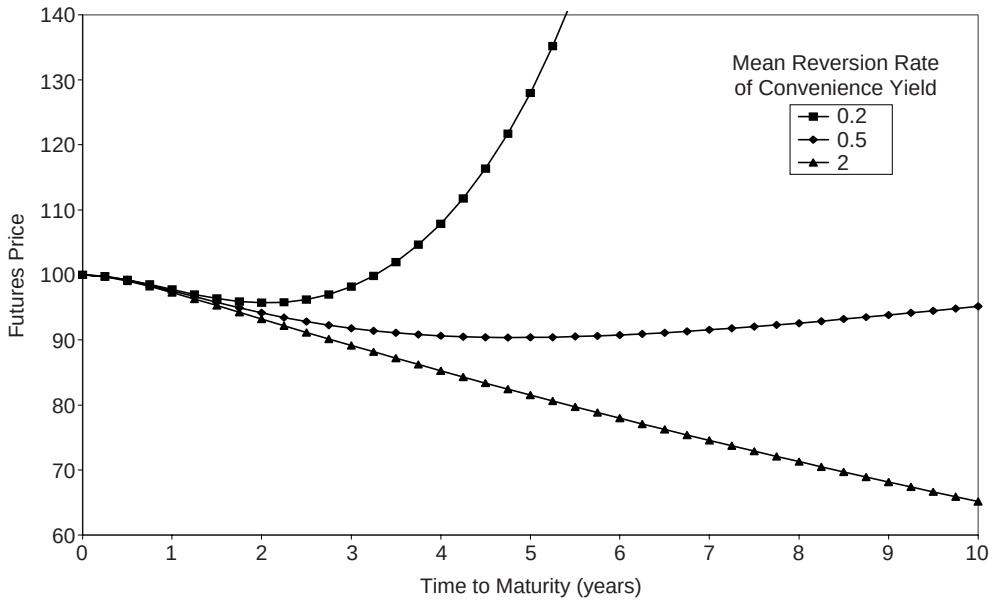


FIGURE 6.6 Futures Prices for the Schwartz Two Factor Model Varying the Speed of Mean Reversion for the Convenience Yield

equation (6.3). These results were originally obtained by Jamshidian and Fein (1990) and Bjerksund (1991).

Figures 6.6 and 6.7 show the sensitivity of the futures pricing formula (6.18) to the speed of mean reversion of the convenience yield and the correlation between the spot price and convenience yield respectively. For both figures we assume $S = 100$ and the following parameters which are roughly typical of the values obtained in Schwartz (1997) for oil; $r = 5\%$, $\delta = 5\%$, $\bar{\delta} = 7\%$, $\alpha_\delta = 2.0$, $\sigma = 30\%$, $\sigma_\delta = 30\%$, $\rho_{S\delta} = 0.8$, $\lambda_\delta = 0$. For Figure 6.6 we have $\alpha_\delta = 0.2, 0.5, 2.0$ and for Figure 6.7 we have $\rho_{S\delta} = 0.2, 0.5, 0.8$.

Figures 6.6 and 6.7 show that the two factor model has a reasonable level of flexibility in the possible shapes of the forward curve (see also Schwartz (1997)). However, the shapes are limited and in particular if the actual curve has more than a single hump or trough then these cannot in general be fitted by the analytical forward curve. In practice, as we have already noted, when pricing options it is not necessary to use the analytical forward curve, the market prices can be used directly.

Application of Itô's lemma to equations (6.17) and (6.18) shows that the volatility of futures contracts are given, in both the Schwartz and Hilliard-Reis framework's by;

$$\sigma_F(t, s) = \sqrt{\sigma^2 + \sigma_\delta^2 H_\delta(t, s)^2 - 2\sigma\sigma_\delta\rho_{S\delta}H_\delta(t, s)} \quad (6.19)$$

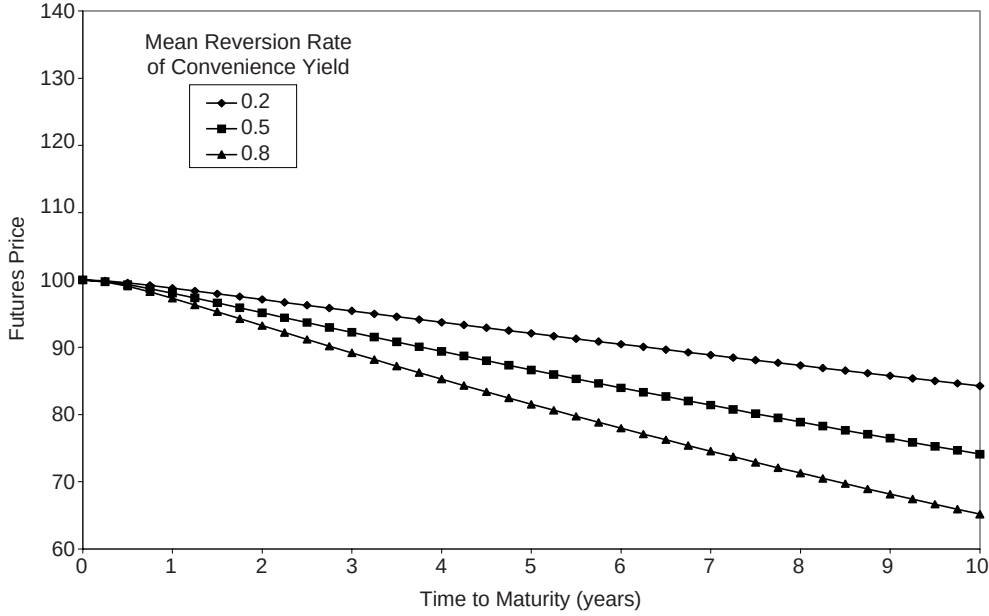


FIGURE 6.7 Futures Prices From Schwartz's Two Factor Model Varying the Correlation Coefficient

It can be seen from equation (6.19) that the volatility of the futures price depends on the volatilities of both the spot asset and the convenience yield, the correlation between the two, the speed of adjustment of the convenience yield, and the maturity of the futures contract. As time to maturity approaches large values, the volatility of futures returns converges to a fixed value, i.e.;

$$\sigma_F(t, \infty) = \sqrt{\sigma^2 + \frac{\sigma_\delta^2}{\alpha_\delta^2} - \frac{2\sigma\sigma_\delta\rho_{S\delta}}{\alpha_\delta}} \quad (6.20)$$

Figure 6.8 shows the sensitivity of the futures volatility formula (6.19) to the correlation coefficient under assumptions for $\rho_{S\delta}$ of 0.75 and 0.9. For both correlations the volatilities attain their long term level, given by equation (6.20) after about 3 years. This figure also shows that these two factor models have reasonably realistic volatility structures with attenuation for short maturities due to the mean reversion but which approach a non-zero level for longer maturities.

Option Pricing

In order to price European futures options for the two factor model we observe from equations (6.17) and (6.18) that the futures price is lognormally distributed and therefore

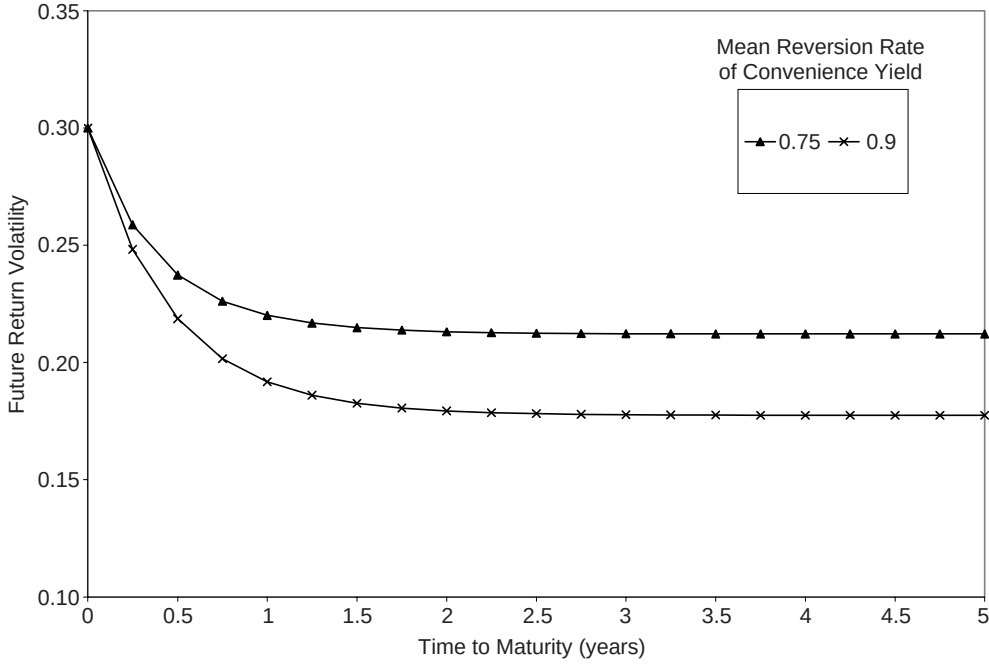


FIGURE 6.8 Futures Return Volatilities for the Schwartz Two Factor Model with Typical Correlations Between the Spot Price and Convenience Yield

option prices are given by a Black type formula but with the appropriate integral of the futures volatility over the life of the option (c.f. equations (6.5) and (6.12)). Clewlow and Strickland (1999b) shows that a European call option, maturing at time T , with strike price K on a futures contract which matures at time s is given by;

$$c(t, T, s) = P(t, T) [F(t, s)N(h) - KN(h - \sqrt{w})] \quad (6.21)$$

$$h = \frac{\ln(F(t, s)/K) + \frac{1}{2}w^2}{\sqrt{w}}$$

$$w^2 = \sigma^2(T - t) - \frac{2\sigma\sigma_\delta\rho_{S\delta}}{\alpha_\delta} \left((T - t) - \frac{(e^{-\alpha_\delta(s-T)} - e^{-\alpha_\delta(s-t)})}{\alpha_\delta} \right)$$

$$+ \frac{\sigma_\delta^2}{\alpha_\delta^2} \left((T - t) - \frac{2}{\alpha_\delta} \left(e^{-\alpha_\delta(s-T)} - e^{-\alpha_\delta(s-t)} \right) + \frac{1}{2\alpha_\delta} \left(e^{-2\alpha_\delta(s-T)} - e^{-2\alpha_\delta(s-t)} \right) \right)$$

By comparison with (6.5), equation (6.21) can be seen as the Black formula with w^2 replacing $\sigma^2(T - t)$. The volatility term in the Black formula depends only on the

volatility of the spot price and the maturity of the option. However, equation (6.21) shows that the volatilities of both the spot price and convenience yield, the correlation coefficient, the speed of mean reversion of the convenience yield, as well as the maturities of both the option and the futures contract are important determinants of the option price.

This formula can be further generalised by replacing the analytical forward prices by the market prices, this can be interpreted as allowing the mean reversion level of the convenience yield to be time dependent. The spot price volatility can also be made time dependent so that seasonal volatility can be modelled – this would require solving the integral of the futures return variance either analytically or numerically as in equation (6.13)⁴.

The Schwartz 1 Factor Approximation

Schwartz (1998) presents a single factor model for energy prices that retains most of the characteristics of his two factor stochastic convenience yield model but is based on the pricing and volatility results of the model represented by equation (6.6). As a consequence, the model only requires the numerical computation corresponding to a typical single-factor model.

His analysis proceeds as follows: the rate of change in the futures price for the two factor model is independent of the initial values of the state variables⁵, i.e.;

$$\left. \frac{1}{F} \frac{\partial F}{\partial s} \right|_{s \rightarrow \infty} = r - \hat{\delta} + \frac{\sigma_{\delta}^2}{2\alpha_{\delta}^2} - \frac{\rho_{S\delta}\sigma\sigma_{\delta}}{\alpha_{\delta}} \quad (6.22)$$

For the one factor model cost of carry model defined by equation (6.3) the rate of change is determined as;

$$\left. \frac{1}{F} \frac{\partial F}{\partial s} \right|_{s \rightarrow \infty} = r - \delta \quad (6.23)$$

Therefore if we now define the convenience yield in equation (6.3) as

$$c = \hat{\delta} - \frac{\sigma_{\delta}^2}{2\alpha_{\delta}^2} + \frac{\rho_{S\delta}\sigma\sigma_{\delta}}{\alpha_{\delta}} \quad (6.24)$$

the one-factor model will have the same rate of change in futures prices as the two-factor model. In order to match the futures prices from the two models' Schwartz defines a shadow spot price dependent on the spot asset and convenience yield, $Z(S, \delta)$, to give the futures price in (6.3) when the constant convenience yield (6.24) is applied;

$$Z(S, \delta) = \lim_{s \rightarrow \infty} e^{-(r-\delta)(s-t)} F(t, s, \delta) = S \exp \left[\frac{c - \delta}{\alpha_{\delta}} - \frac{\sigma_{\delta}^2}{4\alpha_{\delta}^3} \right] \quad (6.25)$$

⁴ The mean reversion rate and volatility of the convenience yield could also be made time dependent.

⁵ The futures price in the two-factor model is given by (6.17).

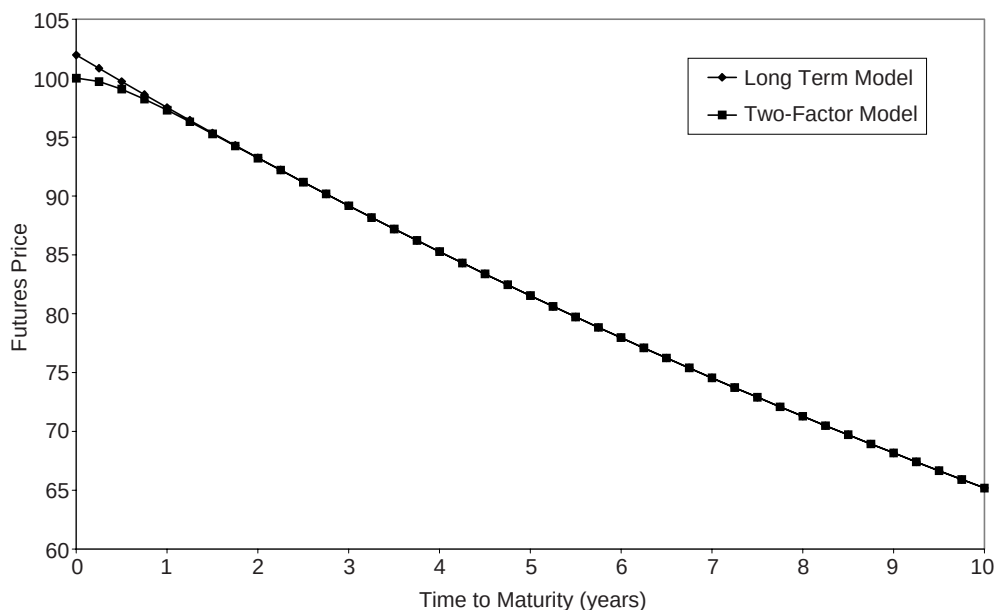


FIGURE 6.9 The Schwartz Two-Factor and Long Term Model Futures Prices

The futures price, for the shadow spot price, is then given by;

$$F(Z, t, s) = Ze^{(r-c)(s-t)} \quad (6.26)$$

Figure 6.9 shows the long term model fitted to a two-factor model with the following parameters; $S = 100$, $r = 5\%$, $\delta = 5\%$, $\bar{\delta} = 7\%$, $\alpha_\delta = 2.0$, $\sigma = 30\%$, $\sigma_\delta = 30\%$, $\rho_{S\delta} = 0.8$, $\lambda = 0$.

Figure 6.9 illustrates that, for this parameter set, the long-term model approximates the two-factor model well for maturities after around 1.5 years. As well as long term prices being very close, an application of Itô's lemma to equation (6.26) shows that these futures price returns have exactly the same volatility as the two-factor model, i.e. equation (6.19). This ad hoc adjustment should therefore produce reasonable prices for simple options on long dated contracts (such as a long dated bond with coupons linked to an energy price) but less well for options with maturities less than 2 years or options which depend on changes in the shape of the forward curve. We would not recommend using ad-hoc approximations such as this for managing portfolios of complex energy derivatives as they tend not to be robust to changing market conditions.

Option Pricing in the Long Term Model

In the long term model, the futures price is lognormally distributed and therefore option prices are given by a Black type formula with the appropriate integral of the futures

volatility over the life of the option. Since the futures volatility in the long term model is identical to the two factor models then the prices of standard European options are given by equation (6.21).

6.4 Three Factor Models

Schwartz (1997) extends his two factor model of the previous section to include stochastic interest rates. In this three factor model the short term rate is assumed to follow the Vasicek (1977) mean reverting process. The full system of defining stochastic differential equations is therefore given by;

$$\begin{aligned} dS &= (r - \delta)Sdt + \sigma Sdz \\ d\delta &= \alpha_\delta(\bar{\delta} - \delta)dt + \sigma_\delta dz_\delta \\ dr &= \alpha_r(\bar{r} - r)dt + \sigma_r dz_r \end{aligned} \quad (6.27)$$

where the correlations between the processes are determined by;

$$\text{cov}(dz, dz_\delta) = \rho_{S\delta}dt \quad \text{cov}(dz, dz_r) = \rho_{Sr}dt \quad \text{cov}(dz_\delta, dz_r) = \rho_{\delta r}dt$$

HR also present a three-factor model in the spirit of Schwartz, but instead of the Vasicek assumption for interest rate evolution they assume that interest rates evolve in an Heath, Jarrow, and Morton (1992) (HJM) world (thus maintaining consistency with the observed yield curve). Under this model instantaneous forward rates are assumed to follow the process;

$$df(t, s) = \alpha(t, s) + \sum_{i=1}^n \sigma_i(t, s)dz_i \quad (6.28)$$

where $f(t, s)$ represents the observed instantaneous forward interest rate curve, $\alpha(t, s)$ is determined by no arbitrage conditions (see HJM) and $\sigma_i(t, s)$ are the associated volatility functions. Equation (6.28) is written in its most general form with n driving sources of randomness. The resulting partial differential equation for contingent claim prices for both Schwartz and HR, is given by;

$$\begin{aligned} \frac{1}{2}C_{SS}S^2\sigma^2 + \frac{1}{2}C_{\delta\delta}\sigma_\delta^2 + \frac{1}{2}C_{rr}\sigma_r^2 + C_{S\delta}S\rho_{S\delta}\sigma\sigma_\delta + C_{Sr}S\rho_{Sr}\sigma\sigma_r + C_{\delta r}S\rho_{\delta r}\sigma_\delta\sigma_r + \\ C_S S(r - \delta) + C_\delta(\alpha_\delta(\bar{\delta} - \delta) - \lambda_\delta\sigma_\delta) + C_r(\mu_r - \lambda_r\sigma_r) + C_t = 0 \end{aligned} \quad (6.29)$$

In the HR framework the risk neutralised drift for the interest rate process is substituted out by a term that does not depend on the market price of risk, λ_r , and so this equation is preference free with regard to interest rate risk. In the Schwartz framework $\mu_r = \alpha_r(\bar{r} - r)$.

Futures and Forward Pricing

Because interest rates are now no longer assumed constant, futures prices will not equal forward prices. It can be shown that the forward price is independent of the short rate

and so is given for the 3 factor model by equation (6.17) for the Schwartz assumptions and equation (6.18) for the HR assumptions. Futures prices, however, are influenced by the short rate and in the Schwartz framework we have futures prices given by;

$$F(t, s) = S \exp \left(-\delta \frac{1 - e^{-\alpha_\delta(s-t)}}{\alpha_\delta} + r \frac{1 - e^{-\alpha_r(s-t)}}{\alpha_r} + D_0(t, s) \right) \quad (6.30)$$

where

$$\begin{aligned} D_0(t, s) = & \frac{(\alpha_\delta \bar{\delta} + \sigma \sigma_\delta \rho_{S\delta})((1 - e^{-\alpha_\delta(s-t)}) - \alpha_\delta(s-t))}{\alpha_\delta^2} \\ & - \frac{\sigma_\delta^2(4(1 - e^{-\alpha_\delta(s-t)}) - (1 - e^{-2\alpha_\delta(s-t)}) - 2\alpha_\delta(s-t))}{4\alpha_\delta^3} \\ & - \frac{(\alpha_r \bar{r} + \sigma \sigma_r \rho_{Sr})((1 - e^{-\alpha_r(s-t)}) - \alpha_r(s-t))}{\alpha_r^2} \\ & - \frac{\sigma_r^2(4(1 - e^{-\alpha_r(s-t)}) - (1 - e^{-2\alpha_r(s-t)}) - 2\alpha_r(s-t))}{4\alpha_r^3} \\ & + \sigma_\delta \sigma_r \rho_{\delta r} \left(\frac{(1 - e^{-\alpha_\delta(s-t)}) + (1 - e^{-\alpha_r(s-t)}) - (1 - e^{-(\alpha_\delta + \alpha_r)(s-t)})}{\alpha_\delta \alpha_r (\alpha_\delta + \alpha_r)} \right. \\ & \left. + \frac{\alpha_\delta^2(1 - e^{-\alpha_r(s-t)}) + \alpha_r^2(1 - e^{-\alpha_\delta(s-t)}) - \alpha_\delta \alpha_r^2(s-t) - \alpha_r \alpha_\delta^2(s-t)}{\alpha_\delta^2 \alpha_r^2 (\alpha_\delta + \alpha_r)} \right) \end{aligned}$$

In order to derive explicit solutions for futures prices, HR restrict the general equation representing the interest rate evolution, (6.28), to a single factor version where volatilities of instantaneous forward rates are equivalent to the Vasicek (1977) model, i.e. $n = 1$ with $\sigma_1(t, s) = \sigma_r e^{-\alpha_r(s-t)}$. Under this restriction futures prices are given by⁶;

$$\begin{aligned} F(t, s) &= SA(t, s) D_1(t, s) D_2(t, s) D_3(t, s) e^{-H_\delta(t, s) \delta} \frac{1}{P(t, s)} \quad (6.31) \\ A(t, s) &= \exp \left[\frac{(H_\delta - (s-t))(\alpha_\delta^2 \alpha_r - \alpha_\delta \lambda_\delta \sigma_\delta - \sigma_\delta^2/2 + \rho_{S\delta} \sigma \sigma_\delta \alpha_\delta)}{\alpha_\delta^2} - \frac{\sigma_\delta^2 H_\delta^2}{4\alpha_\delta} \right] \\ D_1(t, s) &= \exp \left[\frac{\sigma_\delta \sigma_r \rho_{\delta r}}{\alpha_\delta + \alpha_r} \left(\frac{H_r - (s-t)}{\alpha_r} + \frac{H_\delta - (s-t)}{\alpha_\delta} + H_\delta H_r \right) \right] \\ D_2(t, s) &= \exp \left[\frac{\rho_{Sr} \sigma \sigma_r}{\alpha_r} ((s-t) - H_r) \right] \\ D_3(t, s) &= \exp \left[-(H_r - (s-t)) \frac{\sigma_r^2}{\alpha_r^2} - \frac{\sigma_r^2 H_r^2}{2\alpha_r} \right] \\ H_\delta &= H_\delta(t, s) = \frac{1 - e^{-\alpha_\delta(s-t)}}{\alpha_\delta} \\ H_r &= H_r(t, s) = \frac{1 - e^{-\alpha_r(s-t)}}{\alpha_r} \end{aligned}$$

⁶ This restriction of the HJM model is equivalent to the Hull and White (1990) model, often referred to as the “Extended Vasicek” model

Unlike the Schwartz and HR two factor models, the pricing formula (6.31) does not reduce to (6.30) if we assume that the observed term structure is given by $P(t, s) = e^{-r(s-t)}$. This is because under the interest rate assumption in (6.27) the term structure of rates is given by the Vasicek (1977) formulation. Therefore, the prices given by (6.30) and (6.31) are equivalent if we assume that the observed term structure in (6.28) is the Vasicek term structure generated by the Vasicek short rate process in equation (6.27).

Itô's lemma applied to both the defining sets of equations shows that for both the Schwartz and HR models the volatility of the futures price returns is given by;

$$\sigma_F(t, s) = \sqrt{\sigma^2 + \sigma_\delta^2 H_\delta(t, s)^2 + \sigma_r^2 H_r(t, s)^2 - 2\sigma\sigma_\delta\rho_{S\delta}H_\delta(t, s) + 2\sigma\sigma_r\rho_{Sr}H_r(t, s) - 2\sigma\sigma_\delta\rho_{\delta r}H_\delta(t, s)H_r(t, s)} \quad (6.32)$$

The volatility term structure therefore depends on the volatility and speed of mean reversion of the 3 random variables and their correlations, as in the two factor case.

The volatility of futures returns converge to a fixed value as the maturity of the futures contract tends to infinity;

$$\sigma_F(t, \infty) = \sqrt{\sigma^2 + \frac{\sigma_\delta^2}{\alpha_\delta^2} + \frac{\sigma_r^2}{\alpha_r^2} - \frac{2\sigma\sigma_\delta\rho_{S\delta}}{\alpha_\delta} + \frac{2\sigma\sigma_r\rho_{Sr}}{\alpha_r} - \frac{2\sigma\sigma_\delta\rho_{\delta r}}{\alpha_\delta\alpha_r}} \quad (6.33)$$

Option Pricing

Miltersen and Schwartz (1998) (MS) extend the 3 factor Schwartz model of this section to the same as that of HR by assuming that interest rates evolve according to the HJM model. MS derive prices for options on futures under general Gaussian assumptions for the interest rate process. Restricting the volatility function further to be the same as the single factor version in HR, MS derive the following closed form solution for pricing European call options on the futures price;

$$C(t, T, s) = P(t, T) \left[F(t, s) e^a N\left(\frac{\ln(F(t, s)/K) + \alpha + \frac{1}{2}w}{\sqrt{w}}\right) - KN\left(\frac{\ln(F(t, s)/K) + \alpha - \frac{1}{2}w}{\sqrt{w}}\right) \right] \quad (6.34)$$

with

$$w = \sigma(T - t) + 2\sigma \left[\frac{\sigma_r\rho_{Sr}}{\alpha_r} \left(t - \frac{e^{-\alpha_r(s-t)}(e^{\alpha_r(T-t)} - 1)}{\alpha_r} \right) - \frac{\sigma_\delta\rho_{S\delta}}{\alpha_\delta} \left(t - \frac{e^{-\alpha_\delta(s-t)}(e^{\alpha_\delta(T-t)} - 1)}{\alpha_\delta} \right) \right]$$

$$\begin{aligned}
& + \frac{\sigma_\delta^2}{\alpha_\delta^2} \left(t + \frac{e^{-2\alpha_\delta(s-t)}(e^{2\alpha_\delta(T-t)} - 1)}{2\alpha_\delta} - 2 \frac{e^{-\alpha_\delta(s-t)}(e^{\alpha_\delta(T-t)} - 1)}{\alpha_\delta} \right) \\
& + \frac{\sigma_r^2}{\alpha_r^2} \left(t + \frac{e^{-2\alpha_r(s-t)}(e^{2\alpha_r(T-t)} - 1)}{2\alpha_r} - 2 \frac{e^{-\alpha_r(s-t)}(e^{\alpha_r(T-t)} - 1)}{\alpha_r} \right) \\
& - 2 \frac{\sigma_\delta \sigma_r \rho_{\delta r}}{\alpha_\delta \alpha_r} \\
& \left(t - \frac{e^{-\alpha_\delta(s-t)}(e^{\alpha_\delta(T-t)} - 1)}{\alpha_\delta} - \frac{e^{-\alpha_r(s-t)}(e^{\alpha_r(T-t)} - 1)}{\alpha_r} + \frac{e^{-(\alpha_\delta + \alpha_r)(s-t)}(e^{(\alpha_\delta + \alpha_r)(T-t)} - 1)}{\alpha_\delta + \alpha_r} \right) \Big]
\end{aligned}$$

where

$$\begin{aligned}
\alpha = & - \frac{\sigma_r}{\alpha_r} \left(\sigma \rho_{Sr} \left(t - \frac{1 - e^{-\alpha_r(T-t)}}{\alpha_r} \right) \right) \\
& + \frac{\sigma_r}{\alpha_r} \left(t - \frac{e^{-\alpha_r(s-t)}(e^{\alpha_r(T-t)} - 1)}{\alpha_r} - \frac{1 - e^{-\alpha_r(T-t)}}{\alpha_r} + \frac{e^{-\alpha_r(s-t)}(e^{\alpha_r(T-t)} - e^{-\alpha_r(T-t)})}{2\alpha_r} \right) \\
& - \frac{\sigma_\delta \rho_{\delta r}}{\alpha_\delta} \left(t - \frac{e^{-\alpha_\delta(s-t)}(e^{\alpha_\delta(T-t)} - 1)}{\alpha_\delta} - \frac{1 - e^{-\alpha_r(T-t)}}{\alpha_r} + \frac{e^{-\alpha_\delta(s-t)}(e^{\alpha_\delta(T-t)} - e^{-\alpha_r(T-t)})}{\alpha_\delta + \alpha_r} \right)
\end{aligned}$$

Although equation (6.34) looks very complex, comparing it with the previous option pricing equations; (6.5), (6.12) and (6.21), it can be seen to have the same ‘Black’ form but with a more complex expression for the underlying futures return variance over the life of the option.

In practice stochastic interest rates have a relatively small impact on the prices of energy derivatives and interest rates can usually be assumed to be deterministic. These three factor models can therefore be viewed as a useful way of assessing the impact of stochastic interest rates without the need to fully implement the model in a risk management system.

6.5 Choosing a Spot Price Model

The choice of which model to use depends on the nature of the energy markets and derivatives to which it is to be applied. We have shown that all of the spot price models can be made consistent with both the market forward prices and spot volatilities including any seasonal price or volatility patterns.

The key distinguishing feature between the models is the shape of the forward/futures price volatility curve. As we have seen, the prices of options depend on the integral of the underlying forward price variance from its initial value to its value at the option maturity. Since the volatility curve is relatively flat for long maturities then for short maturity options on long maturity forward contracts the Black model could be used.

However for situations in which the integral of the forward price variance is over a non-constant portion of the curve a model which reflects the shape of the curve over the critical region should be used. Thus, for short maturity options on short maturity forwards the Schwartz one factor model can be used. For situations in which the whole volatility curve is important, for example for a large and diverse portfolio of energy contracts, the two factor stochastic convenience yield model is more appropriate. It is usually not necessary to use the three factor model as stochastic interest rates will typically have a relatively minor impact on energy derivatives prices⁷.

The inclusion of jumps into these models leads to the loss of the simple analytical solutions for standard options and therefore the numerical techniques which we discussed in chapter 2 and extend in chapter 7 are needed. In their paper, HR claim to present a quasi-analytical solution for standard options under the three factor model with jumps. However, this solution appears flawed as it is not consistent with the attenuation of the jumps in the case of simple mean reversion.

6.6 Summary

In this chapter we have presented a number of the most popular single and multi-factor spot energy models. For all of these models we have discussed the assumptions and presented the closed form solutions for forward and futures prices and volatilities for constant parameter values, as well as for standard European options. We have indicated how the models are related to each other and how key market data such as observable forward prices and seasonal volatility patterns can be introduced without losing the analytical results. However, for other market factors and for many energy derivatives traded in practice, numerical procedures similar to the techniques introduced in chapter 1 are required to evaluate prices. These techniques are extended to the spot price models of this chapter, in the following chapter.

⁷ A possible exception would be very long dated contracts.